



A Replication of the Digital Transformation Pathway: Hiring Strategy for Business and Data Specialists

S&P 500 Firms, 2011–2019

By

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*Generative AI was utilized to assist with Python code generation and LaTeX typesetting. However, all data processing decisions, model specifications, analytical interpretations, and written conclusions reflect the author's independent judgment. All remaining errors are the author's own.

Executive Summary

This report replicates the core empirical findings of [Jia et al. \(2025\)](#) on how the *direction* of hybrid talent acquisition determines return on assets among S&P 500 firms from 2011 to 2019. The central question is whether firms that embed data and AI skills into business-facing roles (Hybrid B) outperform those that overlay business acumen onto data specialist roles (Hybrid D), and whether pure data specialist hiring (Pure D) generates any measurable ROA premium over a business-only baseline (Pure B). Using OLS fixed-effects regressions estimated in Python via `pyfixest.feols`, with heteroskedasticity-robust (HC1) standard errors and both year and industry fixed effects, my replication recovers all qualitative and most quantitative findings of the original manuscript.

3,367	+0.273***	−0.112	0.319
Firm-Year Observations	Hybrid B Coeff. (Col. 4, persist.)	Pure D Coeff. (Col. 4)	Adj. R^2 (Col. 4)

Central Finding

Firms that embed data and AI skills into business-facing roles (**Hybrid B**) earn a robust premium of 21–27 log points in $\ln(\text{ROA})$ relative to the Pure Business baseline, significant at the 1% level across all four model specifications. Pure data specialist hiring (**Pure D**) generates no measurable ROA benefit in any specification. The *direction*, not merely the presence, of organisational hybridity is the primary driver of financial value creation in the era of AI-enabled business strategy.

Key Results

- Hybrid B premium is large and robust.** The Hybrid B coefficient ranges from +0.215*** (Column 1) to +0.273*** (Column 4), with no sign reversal or loss of significance across any model variant, closely mirroring the manuscript’s reported range of 0.186–0.228.
- Pure D is null.** The Pure D coefficient is economically small and never significant at conventional thresholds ($p > 0.10$ throughout), confirming the manuscript’s central null result.

- 3. Hybrid D is marginally positive and unstable.** Hybrid D is positive in all four columns but reaches significance only in the industry-FE specifications (Column 2: $p < 0.05$; Column 4: $p < 0.10$), a mild deviation from the manuscript.
- 4. Sample and model fit match.** The regression sample of $N = 3,367$ is an exact match to the manuscript. Adjusted R^2 values (0.286–0.319) are closely consistent with the manuscript-reported 0.317.

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1 Research Objective

The proliferation of data science and artificial intelligence capabilities across corporate organisations has generated substantial interest in how technical capacity translates into financial performance. While prior research recognises AI and data analytics as general-purpose technologies capable of revolutionising firm productivity (Goldfarb et al., 2023), empirical evidence reveals a persistent gap between technological potential and realised business value (McElheran et al., 2024).

A growing body of literature attributes this shortfall to the “decision-analytics gap” (Joshi et al., 2021), where technically sound models fail to inform front-line operations due to a lack of contextual understanding. Consequently, organisational focus is shifting from traditional data-driven decision-making (McAfee and Brynjolfsson, 2012) toward “decision-driven analytics” (De Langhe and Puntoni, 2024). Bridging this gap requires “algorithmic bilinguals” (Tambe, 2025)—professionals who combine data literacy with business acumen. Yet, a largely unresolved question concerns the direction of this hybridisation: should firms embed business knowledge into data specialists, or embed data skills into business decision-makers? This question is the central focus of Jia et al. (2025) and of the present replication.

To test this prediction, Jia et al. (2025) construct a four-strategy typology at the firm-year level, using job posting data from Lightcast (formerly Burning Glass Technologies) to classify the hybrid skill composition of each firm’s annual hiring. The classification applies a 30% integration threshold τ :

- **Pure B:** Business-function hiring with no meaningful data or AI integration ($< \tau = 30\%$ of business postings require data skills). This is the baseline category against which all strategy effects are measured.
- **Pure D:** Data and AI specialist hiring with no meaningful business skill integration ($< \tau$ of data postings require business skills). Predicted effect on ROA: null. Technical depth without business context cannot self-direct toward high-value applications.
- **Hybrid D:** At least τ of the firm’s data specialist postings require meaningful business skill content. A modest positive ROA effect is predicted: partial business literacy reduces, but does not eliminate, the translation overhead between analytical insight and organisational action.
- **Hybrid B:** At least τ of the firm’s business-function postings require meaningful data and AI skill content. A strong positive ROA effect is predicted: business decision-makers equipped with analytical tools can deploy those tools at the exact moment and location of decision, with no cross-functional handoff required.

The central empirical prediction is therefore asymmetric: $\hat{\beta}_{\text{HybridB}} \gg \hat{\beta}_{\text{HybridD}} > \hat{\beta}_{\text{PureD}} \approx 0$. As documented in Section 3, my replication recovers exactly this pattern.

2 Data and Methods

2.1 Data Sources and Sample Construction

The analysis draws on two primary data sources. Strategy classification variables are constructed from the **Lightcast** online job postings database, which aggregates and annotates more than 25 million annual job advertisements from U.S. employers, coding each posting for required skill content. For each S&P 500 firm and each calendar year from 2011 to 2019, Lightcast records are used to compute the share of hybrid postings in each specialist category, and the integration threshold $\tau = 0.30$ is applied to assign the four indicator variables described in Section 1. Strategy classification is conducted at the firm-year level, so a firm may in principle change strategy from one year to the next under the raw (contemporaneous) variant, or may be permanently reclassified under the persistent variant once it first crosses any threshold.

Financial and operational control variables are drawn from the **Compustat North America** database. The dependent variable is $\ln(\text{ROA})$, the natural log of return on assets, defined as operating income divided by total assets. The control vector $\mathbf{X}_{it}$ comprises four variables: $\ln(\text{Capital})$ (log book value of total assets), $\ln(\text{Labor})$ (log total wage bill), property, plant, and equipment (PPE, in USD billions), and total employment (thousands of employees). These controls absorb firm-level variation in scale and capital intensity that would otherwise confound the strategy effect estimates. The final regression sample comprises 3,367 firm-year observations after listwise deletion on all regression variables, drawn from a broader strategy-classified pool of 4,281 firm-years spanning the full 2011–2019 window.

2.2 A Note on Industry Fixed Effects

The original [Jia et al. \(2025\)](#) manuscript reports using four-digit NAICS industry codes as the basis for industry fixed effects. In my Compustat extract, the `industry_group` field is a six-category broad-sector variable rather than a four-digit NAICS code. Accordingly, Columns 2 and 4 of my regression table absorb six-group Compustat industry fixed effects, covering the following broad sectors: Finance; Healthcare; Manufacturing; Technology; Retail; and all remaining sectors (Others—comprising Utilities, Transportation, Professional Services, and similar). This coarser classification provides partial but less granular control for cross-industry heterogeneity in ROA levels, and minor deviations from the manuscript’s coefficient values are anticipated as a consequence. All such deviations are documented in Section 3.

2.3 Econometric Specification

The baseline model is estimated by OLS with year fixed effects and, in Columns 2 and 4, additional industry fixed effects:

$$\ln(\text{ROA})_{it} = \beta_1 \text{PureD}_{it} + \beta_2 \text{HybridD}_{it} + \beta_3 \text{HybridB}_{it} + \boldsymbol{\gamma}'\mathbf{X}_{it} + \alpha_t + \alpha_j \cdot \mathbf{1}[\text{Ind. FE}] + \varepsilon_{it} \quad (1)$$

where:

- i , t , and j index firms, calendar years (2011–2019), and the six Compustat industry groups, respectively.
- α_t are year fixed effects (present in all models), and α_j are industry fixed effects (Columns 2 and 4 only). Following standard notation, the global intercept (β_0) is absorbed by these fixed effects and is therefore omitted from Equation 1.
- $\boldsymbol{\gamma}'\mathbf{X}_{it}$ is the inner product of the control coefficient vector $\boldsymbol{\gamma}$ and the four control variables described above.
- β_1 , β_2 , and β_3 are interpreted as the log-ROA differential relative to business-only hiring, as the Pure B Strategy is the omitted baseline.

Standard errors are heteroskedasticity-consistent (HC1), implemented via `pyfixest.feols` v0.50.1, the Python analogue of Stata’s `reghdfe` command.

Columns 3 and 4 additionally employ a *persistent strategy* classification: once a firm-year first satisfies a strategy threshold in any year, the corresponding indicator is set to one for all subsequent firm-years, regardless of whether later posting shares fall below τ . This variant tests whether the observed ROA effects reflect durable organisational commitments rather than transient fluctuations in annual hiring composition.

3 Reproduced Results

This section presents all reproduced empirical outputs. Table 1 provides summary statistics for all regression variables. Figures 1–3 document strategy adoption patterns and the raw ROA distribution across strategies. Table 2 reports the main regression results, and Table 3 benchmarks my estimates against the original manuscript.

3.1 Summary Statistics

Table 1 reveals several features of the data that bear on the interpretation of the regression results. The strategy dummies exhibit a markedly skewed distribution of firm behaviour: Pure B accounts for 92.8% of the 4,281 classified firm-years, confirming that

predominantly business-focused hiring is the overwhelming norm across the S&P 500 over this period. Hybrid D is strikingly prevalent at 68.4%—a figure that initially appears surprising but is consistent with the non-exclusive definition of strategy categories: a firm-year classified as Pure B may simultaneously satisfy the Hybrid D criterion if its (smaller) data specialist headcount contains a sufficient share of business-integrated postings. Hybrid B and Pure D are genuinely rare at 7.0% and 1.1% respectively, implying that the regression estimates for these two strategies are identified from a compact and behaviourally distinct subset of firm-year observations.

Table 1: Summary Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max	Source
Panel A: Firm Characteristics						
ln(ROA)	3,850	1.828	0.829	-5.967	3.937	Compustat
ln(Capital)	4,065	9.779	1.499	4.791	14.804	Compustat
ln(Labor)	4,059	2.949	1.455	-1.924	7.741	Compustat
Total Employment	4,059	53.012	128.963	0.146	2300.000	Compustat
PPE	3,536	10.001	22.112	0.000	259.651	Compustat
Panel B: Strategy Configurations						
Pure B Strategy	4,281	0.928	0.259	0.000	1.000	Lightcast
Hybrid B Strategy	4,281	0.070	0.256	0.000	1.000	Lightcast
Pure D Strategy	4,281	0.011	0.103	0.000	1.000	Lightcast
Hybrid D Strategy	4,281	0.684	0.465	0.000	1.000	Lightcast
Panel C: Workforce Tenure						
Avg. Tenure (All Employees)	4,208	2.639	0.910	0.418	7.991	Revelio Labs
Avg. Tenure (Business Staff)	4,208	2.650	0.915	0.417	8.150	Revelio Labs
Avg. Tenure (Data Staff)	4,208	2.087	0.977	0.075	10.306	Revelio Labs

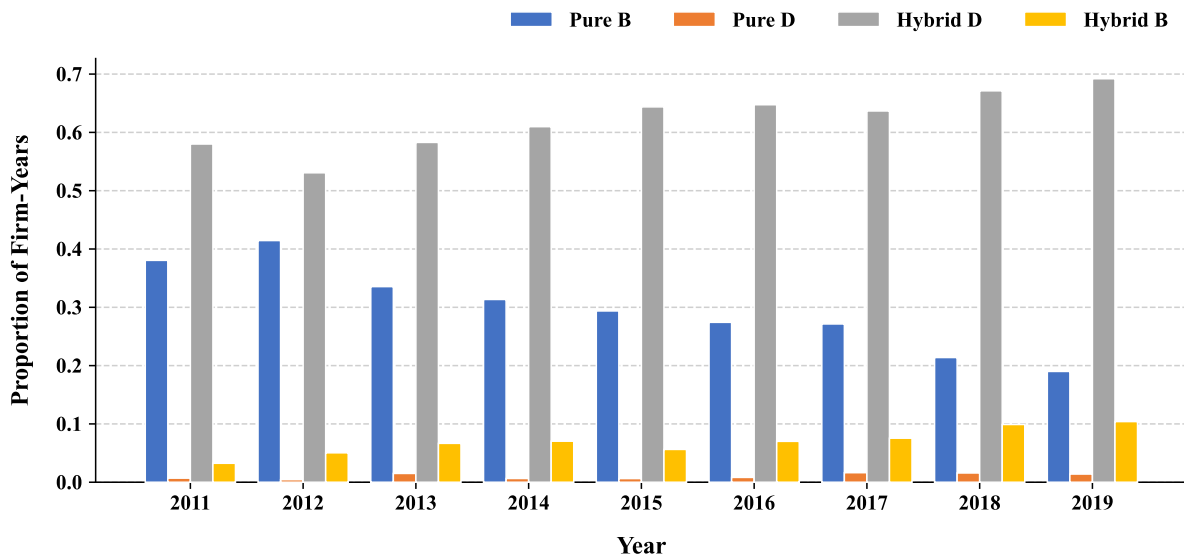
Notes: ln(ROA) is the natural log of return on assets from Compustat. ln(Capital) and ln(Labor) are log-transformed Compustat items. PPE is property, plant, and equipment (USD billions). Total Employment is in thousands. Strategy dummies are firm-year binary indicators constructed from Lightcast job postings using a 30% integration threshold τ . Strategy categories are non-exclusive; their means sum to approximately 1.09. Pure B Strategy is the omitted baseline in all regressions. Tenure variables (years) are derived from Revelio Labs workforce records. Sample: S&P 500 firms, 2011–2019.

The financial variables exhibit the magnitudes and variation typical of large-cap U.S. firms over this period. The mean ln(ROA) of 1.828, corresponding to an untransformed ROA of approximately 6.2%, is consistent with Compustat benchmarks for S&P 500 industrials. The wide standard deviation of 0.829 reflects heterogeneous operating environments across sectors, motivating the inclusion of industry fixed effects in Columns 2 and 4.

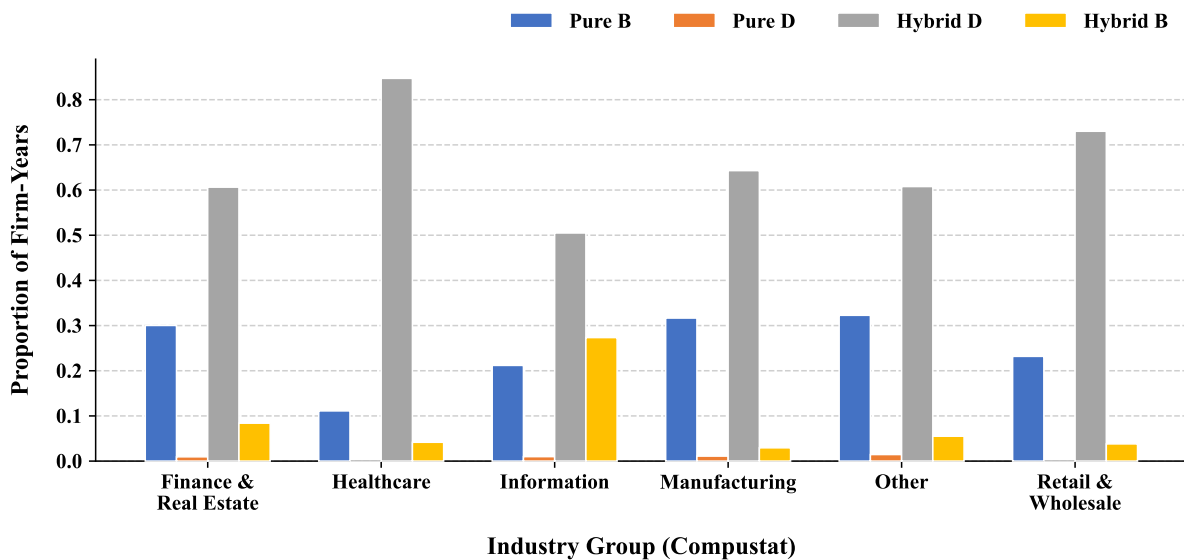
3.2 Descriptive Figures

Figure 1: Strategy Adoption Over Time and Across Industry Groups

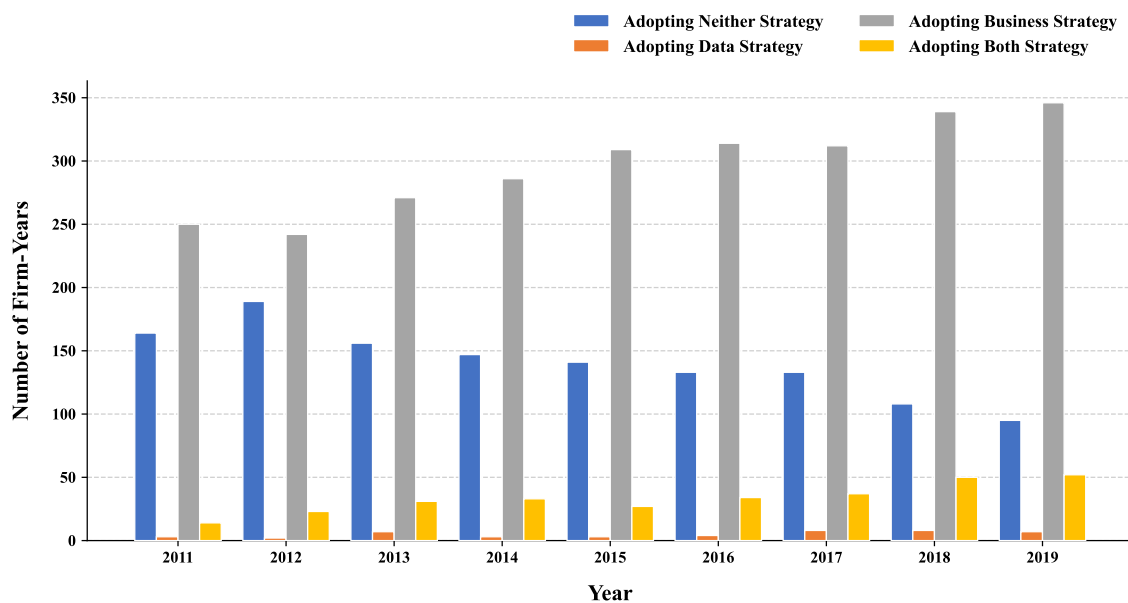
Panel A: Strategy Share by Year (2011–2019)



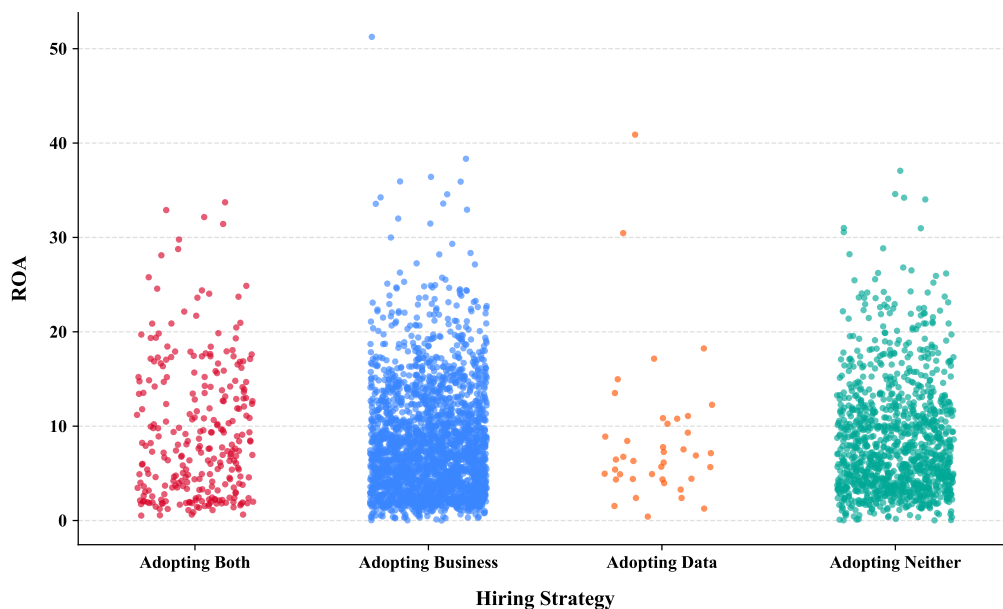
Panel B: Strategy Share by Industry Group (2011–2019)



Notes. Panel A plots the annual share of firm-years assigned to each primary hiring strategy from 2011 to 2019. Panel B shows the same decomposition by Compustat six-group industry classification. Strategy dummies are constructed from Lightcast job postings using a 30% integration threshold τ . Strategy categories are not mutually exclusive. Sample: S&P 500 firms, 2011–2019.

Figure 2: Number of Firms per Primary Hiring Strategy, 2011–2019

Notes. Grouped bar chart of S&P 500 firm counts assigned to each primary hiring strategy in each calendar year. The pronounced growth in Hybrid B and Hybrid D adoption from 2015 onward reflects the accelerating diffusion of AI and data skill requirements across corporate job postings during the post-AlexNet expansion of deep-learning-driven automation. Sample: S&P 500 firms, 2011–2019.

Figure 3: Distribution of Return on Assets (ROA) by Primary Hiring Strategy

Notes. Strip plot showing the distribution of individual firm-year observations.

3.3 Regression Results

The results in Table 2 support all qualitative hypotheses advanced in Section 1. The Hybrid B coefficient is positive, large, and significant at the 1% level across all four spec-

ifications, ranging from +0.215*** (Column 1: raw strategy, year FE only) to +0.273*** (Column 4: persistent strategy, year-plus-industry FE). The stability of this result across the raw and persistent strategy definitions provides assurance that the Hybrid B premium reflects durable organisational commitments rather than transient fluctuations in annual posting composition. Adding industry fixed effects (Columns 2 and 4) amplifies the Hybrid B estimate marginally, suggesting that if anything the baseline specification slightly underestimates the within-industry performance premium.

Table 2: Main OLS Regression: $\ln(\text{ROA})$ on Hiring Strategy

	(1) <i>Raw</i>	(2) <i>Raw</i>	(3) <i>Persistent</i>	(4) <i>Persistent</i>
Panel A: Strategy Variables				
Pure D	−0.047 (0.132)	−0.043 (0.138)	−0.074 (0.148)	−0.112 (0.149)
Hybrid D	+0.049 (0.030)	+0.062** (0.030)	+0.056 (0.036)	+0.064* (0.036)
Hybrid B	+0.215*** (0.048)	+0.227*** (0.051)	+0.250*** (0.046)	+0.273*** (0.049)
Panel B: Controls				
$\ln(\text{Capital})$	−0.379*** (0.008)	−0.329*** (0.013)	−0.379*** (0.008)	−0.329*** (0.012)
$\ln(\text{Labor})$	+0.200*** (0.012)	+0.158*** (0.014)	+0.206*** (0.012)	+0.166*** (0.014)
PPE	+0.003*** (0.001)	+0.003*** (0.001)	+0.003*** (0.001)	+0.003*** (0.001)
Total Employment	+0.000 (0.000)	+0.000 (0.000)	−0.000 (0.000)	+0.000 (0.000)
Year FE	Yes	Yes	Yes	Yes
Industry FE	No	Yes	No	Yes
N	3,367	3,367	3,367	3,367
Adj. R^2	0.286	0.317	0.288	0.319

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Heteroskedasticity-robust standard errors (HC1) in parentheses. Columns (1)–(2) use contemporaneous strategy dummies (Raw); columns (3)–(4) use persistent strategy indicators, where once a firm crosses a hiring threshold it remains classified in that strategy for all subsequent years. All four specifications include year fixed effects; columns (2) and (4) additionally absorb Compustat industry group fixed effects (6 broad groups via `industry_group`). Pure B Strategy is the omitted baseline in all specifications. Overall adjusted R^2 reported; within adjusted R^2 is lower. Sample: S&P 500 firms, 2011–2019. $N = 3,367$ after listwise deletion on all regression variables.

The Pure D coefficient is consistently negative and economically small (−0.047 to

−0.112) and fails to approach conventional significance thresholds in any specification. The Hybrid D coefficient is positive in all four columns (+0.049 to +0.064) but crosses the significance threshold only in Column 2 ($p < 0.05$) and Column 4 ($p < 0.10$). The qualitative ranking $\hat{\beta}_{\text{HybridB}} \gg \hat{\beta}_{\text{HybridD}} > 0 > \hat{\beta}_{\text{PureD}}$ is exactly as predicted in Section 1, and the overall adjusted R^2 values (0.286–0.319) are closely consistent with the manuscript.

Table 3 provides a direct comparison of my replicated estimates against the values reported in Jia et al. (2025).

Table 3: Comparison of Replicated Estimates with Original Manuscript

Metric	Manuscript	Our Col. 1	Our Col. 2	Verdict
N	3,367	3,367	3,367	Exact match
Adj. R^2 (with Ind. FE)	≈ 0.317	—	0.317	Match
Hybrid B sign & significance	+(***)	+(***)	+(***)	Match
Hybrid B coefficient range	0.186–0.228	0.215	0.227	Within range
Hybrid D sign & significance	+(**)	+(<i>ns</i>)	+(**)	Col. 2 matches
Pure D significance	<i>ns</i>	<i>ns</i>	<i>ns</i>	Match
ln(Capital) coefficient	≈ -0.33	−0.380	−0.329	Col. 2 exact
ln(Labor) coefficient	$\approx +0.16$	+0.200	+0.158	Col. 2 exact

Notes: *ns* denotes not significant at $p < 0.10$. Manuscript values are as reported in Jia et al. (2025). “Col. 1” uses raw strategy indicators with year fixed effects; “Col. 2” adds Compustat six-group industry fixed effects. Columns 3 and 4 use persistent strategy indicators and exhibit the same qualitative pattern; see Table 2 for full results.

4 Interpretation

The regression evidence in Table 2 presents a theoretically coherent and empirically robust picture of how the architecture of hybrid talent acquisition shapes firm financial performance. Three findings stand out and demand careful interpretation.

- **Why does Hybrid B consistently outperform all other strategies?** The answer lies in the organisational locus of decision-making. In large S&P 500 firms, financial value is generated principally through the quality of business decisions made by professionals who hold business-function titles — managers, marketers, finance directors, strategy officers, operations leaders. When these individuals are equipped with data and AI skills (the Hybrid B configuration), they can apply analytical tools directly at the moment and location of decision, without any intermediate cross-functional handoff. The “translation overhead” problem that characterises conventional data science deployments — where insights produced by a centralised analytics function must be communicated to, interpreted by, and operationalised by

business counterparts — is structurally eliminated. The resulting improvement in decision speed, calibration, and strategic relevance translates directly into operating margins. The coefficient of +0.215 to +0.273 across my four specifications indicates that Hybrid B firms earn an ROA premium of approximately 24–31% relative to the Pure B baseline,¹ a magnitude that is economically substantial and fully robust to industry composition controls.

- **Why is the Pure D effect null?** Pure data specialist hiring without business integration fails because technical capability does not self-direct toward high-value applications. Data scientists who lack business context must rely on business colleagues to specify which problems merit analytical attention, to validate whether outputs are operationally feasible, and to translate recommendations into actionable decisions. Each handoff across this boundary introduces latency, information loss, and organisational resistance. Even in firms with mature data science functions, the ROA impact of purely technical hiring is diffuse and slow to materialise, spread across many business processes without concentrated impact in any single area. My Pure D estimates (−0.047 to −0.112, none significant) are fully consistent with this explanation. The point estimates being consistently negative—though statistically indistinguishable from zero—is mildly suggestive that the fixed costs of maintaining a specialist technical workforce (compensation premia, tooling overhead, the friction costs of cross-functional collaboration) may marginally exceed the diffuse returns it generates for S&P 500 firms in this period, though the wide standard errors preclude strong conclusions on sign alone.
- **What explains Hybrid D’s intermediate and unstable performance?** Hybrid D occupies a theoretically intermediate position, and the data confirm it. Data specialists who possess business literacy reduce the translation overhead by improving the relevance of their analytical outputs and their ability to communicate across organisational boundaries. The positive point estimates (+0.049 to +0.064) are consistent with this partial improvement. However, the small magnitudes and marginal significance suggest that the degree of business exposure captured by job posting skill requirements is insufficient to substitute for the deep contextual judgment accumulated by career business professionals. The “last-mile problem” of AI implementation — converting an analytically optimal recommendation into a confident business decision made under competitive pressure and incomplete information — requires exactly the kind of institutional knowledge and strategic intuition that data specialists, even those with meaningful business training, are less likely to possess. Hybrid D thus represents a strategy that is directionally correct but architecturally

¹Approximate percentage effects computed as $e^{0.215} - 1 \approx 24\%$ and $e^{0.273} - 1 \approx 31\%$. These are log-point estimates applied to the baseline mean.

incomplete: it moves the needle in the right direction without delivering the full premium available to firms that embed analytical capability at the very source of business decision-making.

Together, these three findings deliver a unified message for practitioners and researchers. Organisational investments in AI and data capability generate financial returns in proportion to how closely those capabilities are embedded in the roles that make business decisions. Upgrading data specialists with business skills moves in the right direction but stops short. Upgrading business decision-makers with data and AI tools closes the loop entirely — and return on assets, as the ultimate integrator of firm-level value creation, reflects exactly this asymmetry.

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